

Integrated Sensing and Backscatter Communication (ISABC): Toward Self-Sustainable Autonomous IoT Networks

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Abstract—Backscatter communication has emerged as a promising enabler for battery-free and low-complexity Internet-of-Things (IoT) devices, while advances in radar sensing and millimeter-wave systems have expanded its capabilities beyond identification toward localization, monitoring, and environmental awareness. More recently, these trends have converged into integrated sensing and backscatter communication (ISABC), where the same reflected waveform supports both data transmission and sensing. This article reviews the foundations and recent progress of ISABC with emphasis on frequency-modulated continuous-wave (FMCW) radar-based implementations, which is meaningful to clarify how range estimation, tag identification, and data transfer can be jointly realized within a unified passive architecture. The article highlights both the opportunities and the current limitations of ISABC, and argues that future systems will evolve toward spatially aware, waveform-coordinated, and electromagnetically engineered backscatter platforms for battery-free sensing and 6G integrated wireless intelligence.

Index Terms—backscatter communication, ISAC, FMCW radar, IoT, metasurface.

I. INTRODUCTION

Over the past decades, backscatter communication has emerged as one of the most energy-efficient wireless communication technologies for Internet-of-Things (IoT) and wireless sensor networks [1]–[3]. Unlike conventional wireless communication systems that actively generate radio frequency (RF) signals using oscillators and power amplifiers, backscatter communication enables passive devices to transmit information by modulating and reflecting incident electromagnetic waves [2]. This mechanism significantly reduces power consumption and hardware complexity, making backscatter communication particularly attractive for battery-free sensors, massive IoT deployments, and low-cost wireless devices [1].

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Early backscatter systems were primarily developed for radio-frequency identification (RFID) systems, where passive tags communicate with readers by modulating their radar cross section (RCS). Over time, backscatter technology has evolved beyond RFID into wireless sensing, localization, and environmental monitoring systems. Recent advances in millimeter-wave radar and passive backscatter tags have enabled high-resolution localization, motion tracking, and infrastructure sensing applications [4]–[8].

More recently, researchers have begun integrating sensing and communication functionalities into unified backscatter systems. This concept, often referred to as Integrated Sensing and Backscatter Communication (ISABC), enables simultaneous data communication and environmental sensing using the same backscatter hardware platform. Several works have demonstrated integrated radar backscatter communication systems using frequency-modulated continuous-wave (FMCW) radar signals and passive backscatter tags [9]–[12].

As shown in Fig. 1(a), this concept is closely related to the recently emerging concept of Integrated Sensing and Communication (ISAC), which is considered a key technology for future 6G wireless systems [10]. However, ISABC fundamentally differs from conventional ISAC systems [10]. In traditional ISAC systems, both sensing and communication are performed using active transceivers that generate RF signals and share waveform, spectrum, and hardware resources [10]. In contrast, ISABC systems rely on passive backscatter tags that do not generate RF signals but instead modulate and reflect incident radar waveforms, as shown in Fig. 1(b) [9]. As a result, ISABC systems enable ultra-low-power integrated sensing and communication, making them particularly suitable for battery-free IoT sensors, smart environments, and large-scale distributed sensing network, as shown in Fig. 2 [1], [13], [14].

The evolution of backscatter technology can be viewed in several stages: RFID-based identification systems, backscatter communication networks, backscatter sensing systems, and finally ISABC systems. This evolution reflects the increasing demand for low-power wireless communication and sensing technologies in IoT, smart city, and autonomous system applications [1], [3], [10].

Despite significant progress in backscatter communication, sensing backscatter systems and communication backscatter systems have historically been developed independently [3]. Sensing backscatter systems primarily focus on localization, tracking, and environmental sensing using radar signals and backscatter tags [4]–[8]. In contrast, communication backscat-

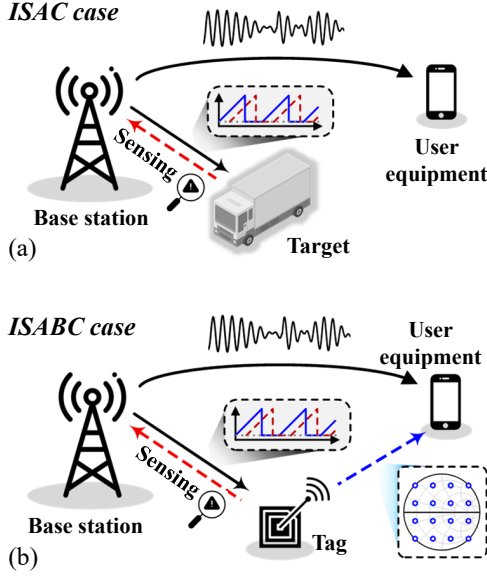


Fig. 1. System model of (a) conventional integrated sensing and communications (ISAC) and (b) developed integrated sensing and backscattering communication (ISABC).

ter systems focus on data transmission, modulation techniques, and network architectures for passive communication [15]–[26]. However, both types of systems rely on the same physical principle—modulation of reflected electromagnetic waves through impedance switching [2]. This observation naturally leads to the integration of sensing and communication into a unified backscatter system, which forms the basis of ISABC.

In recent years, several research efforts have demonstrated ISABC systems leveraging FMCW radar waveforms, chirp-based signaling, and millimeter-wave platforms, enabling a unified architecture for simultaneous communication and sensing [9]–[11]. These systems exploit passive backscatter tags to jointly perform range, velocity, and angle estimation while embedding information modulation, as evidenced by recent works on high-precision localization and beam-aware backscatter sensing [4], [6]–[8]. Such integration opens new opportunities for smart environments, autonomous systems, and large-scale IoT deployments, where battery-free operation and minimal maintenance are critical requirements.

Despite recent advances, current ISABC systems still suffer from limited communication throughput, severe double path loss, limited spatial multiplexing capability, and coverage degradation in blockage environments. Furthermore, since sensing and communication often share the same waveform and hardware platform, a fundamental trade-off exists between sensing accuracy and communication data rate.

Future ISABC systems are therefore expected to evolve toward multi-antenna backscatter systems, distributed sensing networks, metasurface-based backscatter platforms, and networked radar–communication architectures. Multi-input multi-output (MIMO) backscatter systems can improve spectral efficiency and communication range via spatial multiplexing and beamforming [27], [28], while distributed multi-tag networks enable large-scale environmental sensing and

cooperative localization [25], [29]. In addition, reconfigurable intelligent surfaces and metasurface-based backscatter structures can improve coverage in blockage environments and enhance link budgets by controlling electromagnetic wave propagation [30]–[33]. These technologies are expected to constitute foundational enablers of sensing-integrated wireless architectures in future IoT and 6G networks, where joint spectrum utilization, hardware sharing, and passive operation are anticipated to become core system requirements.

The remainder of this article is organized as follows. Section II introduces the fundamentals of backscatter communication and signal models, including impedance modulation and backscatter link analysis. Section III reviews sensing backscatter systems and millimeter-wave backscatter localization techniques. Section IV discusses ISABC systems, including FMCW radar backscatter and joint sensing and communication architectures. Section V reviews backscatter modulators, modulation techniques, and MIMO backscatter systems in the context of integrated sensing and communication. Finally, Section VI concludes the article and discusses future research directions for Integrated Sensing and Backscatter Communication systems.

II. BACKGROUND: BACKSCATTER AND FMCW RADAR SIGNAL MODELS

ISABC systems combine backscatter communication and radar sensing into a unified signal processing framework. To understand ISABC systems, it is necessary to review the signal models of backscatter communication and FMCW radar systems. This section introduces the signal model of backscatter system, the FMCW radar ranging principle, and a unified signal model that explains how sensing and communication can be performed simultaneously in ISABC systems.

A. Backscatter Signal Model

As shown in Fig. 3, backscatter signal operates by modulating and reflecting an incident electromagnetic wave instead of generating a new RF signal, where the tag modulates an externally supplied carrier by changing its reflection coefficient [34]. The carrier signal from the reader radiates a non-modulated continuous wave (CW) at a carrier frequency at f_c . The backscatter tag connected to a backscatter modulator alters the tag impedance for the incident wave. The frequency of modulated backscattered field is located in the sidebands of the backscattered field at $f_c \pm (n/T)$, where T represents the symbol period. This process is commonly referred to as load modulation [35].

The reflection coefficient of a tag antenna is defined as

$$\Gamma = \frac{Z_L - Z_A^*}{Z_L + Z_A} \quad (1)$$

where Z_L is the load impedance connected to the antenna and Z_A is the antenna impedance. By switching the load impedance between different values, the reflection coefficient becomes time-varying and can be used to encode information.

Assuming that the reader transmits a continuous-wave signal, the transmitted signal can be expressed as

$$s(t) = A \cos(2\pi f_c t) \quad (2)$$

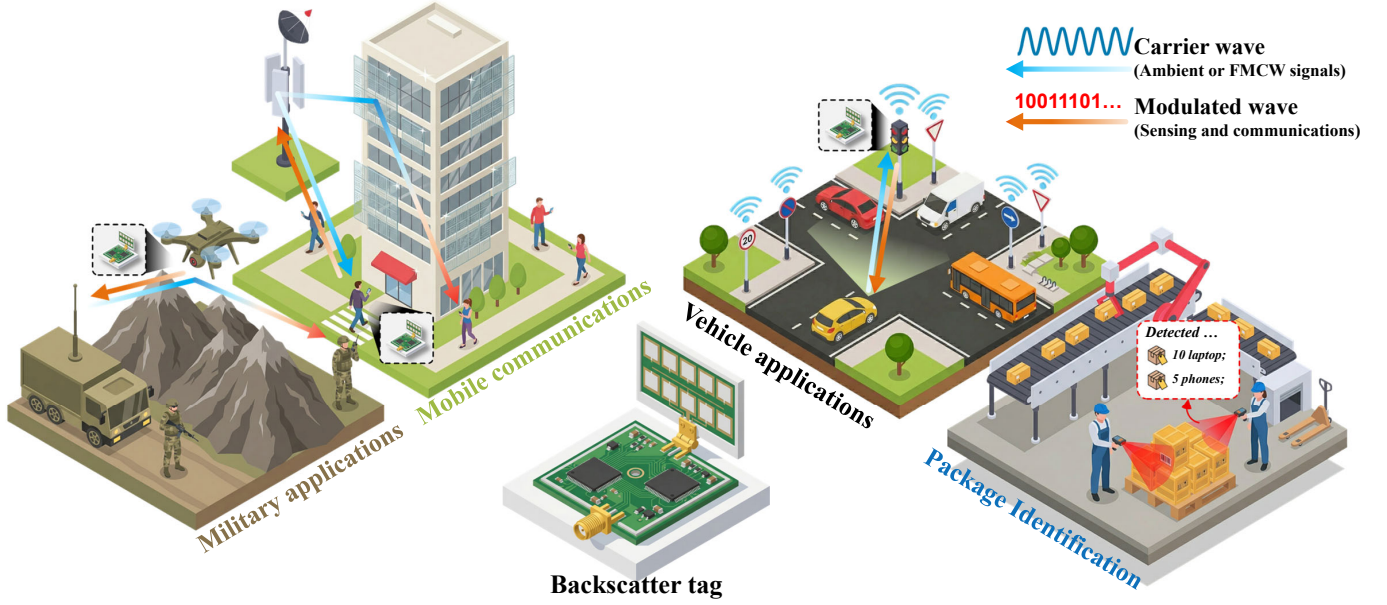


Fig. 2. ISABC systems provide ultra-low-power integrated sensing and communication for several applications proper for ultra-low-power communication and large-scale distributed sensing network.

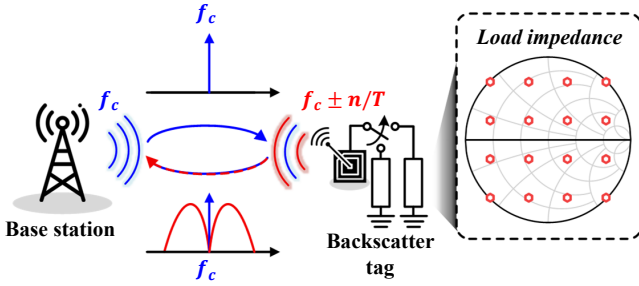


Fig. 3. Description of backscatter communication between a tag and a reader. The tag is designed to modulate its impedance among N different states.

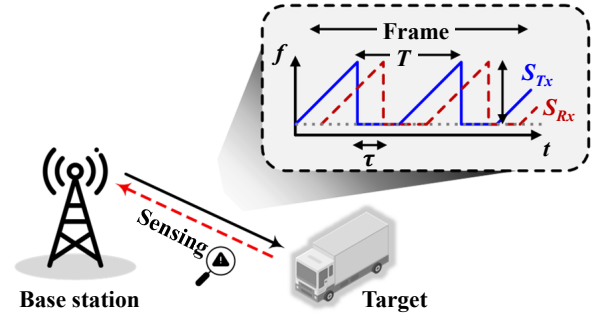


Fig. 4. Description of the sensing scenario using FMCW radar.

The signal propagates from the reader to the tag and then back to the reader after reflection. Therefore, backscatter communication experiences a two-way propagation channel. The received signal at the reader can be written as

$$r(t) = h_{rt}h_{tr}\Gamma(t)s(t) \quad (3)$$

where h_{rt} and h_{tr} represent the reader-to-tag and tag-to-reader channels, respectively. This equation shows that backscatter signal suffers from double path loss due to the two-way propagation between the reader and the tag, which is one of the main limitations of backscatter system range.

B. FMCW Radar Signal Model

In ISABC systems, sensing is typically performed using FMCW radar. FMCW radar transmits a chirp signal whose frequency increases linearly over time. As shown in Fig. 4, the range of a target can be estimated by measuring the frequency difference between the transmitted and received signals [10].

The transmitted FMCW chirp signal can be expressed as

$$s(t) = A \cos \left(2\pi \left(f_c t + \frac{K}{2} t^2 \right) \right) \quad (4)$$

where f_c is the starting frequency and K is the chirp slope defined as

$$K = \frac{B}{T} \quad (5)$$

where B is the frequency bandwidth and T is the chirp duration. If the signal is reflected from a target located at distance R , the round-trip delay is

$$\tau = \frac{2R}{c} \quad (6)$$

where c is the speed of light. After mixing the transmitted (S_{tx}) and received (S_{rx}) signals, the beat frequency is obtained as

$$f_b = K\tau = \frac{2RK}{c} \quad (7)$$

Thus, the target range can be estimated from the beat frequency as

$$R = \frac{cf_b}{2K} \quad (8)$$

This is the fundamental ranging principle of FMCW radar systems.

C. ISABC Signal Model

In ISABC systems, the FMCW radar waveform transmitted from the reader is reflected by passive or semi-passive backscatter tags. The tag does not generate an RF carrier by itself; instead, it embeds communication data by changing its time-varying reflection coefficient, denoted by $\Gamma(t)$. Therefore, the received signal at the reader contains both radar sensing information and backscatter communication data.

The signal backscattered from a tag can be expressed as

$$r(t) = h_{rt}h_{tr}\Gamma(t) \cos\left(2\pi\left[f_c(t-\tau) + \frac{K}{2}(t-\tau)^2\right]\right) \quad (9)$$

where h_{tr} and h_{rt} denote the forward and backward propagation channels, respectively, $\Gamma(t)$ is the time-varying reflection coefficient of the tag, and $\tau = 2R/c$ is the round-trip delay for a tag located at range R .

At the reader, the received signal is mixed with a copy of the transmitted FMCW signal and then low-pass filtered. The high-frequency mixing product around $2f_c$ is removed, and the remaining dechirped beat signal can be written as

$$r_b(t) = A\Gamma(t) \cos(2\pi f_b t + \phi_0), \quad (10)$$

where A includes the channel and scattering amplitudes, ϕ_0 is a constant phase term, and f_b is the FMCW range beat frequency.

Assuming that the tag has a radial velocity v , the reflected signal also experiences a Doppler frequency shift. The Doppler frequency is defined as

$$f_d = \frac{2v}{\lambda} = \frac{2vf_c}{c} \quad (11)$$

where $\lambda = c/f_c$ is the wavelength. Under the slow-motion approximation within one chirp, the Doppler shift can be included in the dechirped signal as

$$r_b(t) = A\Gamma(t) \cos(2\pi(f_b + f_d)t + \phi_0). \quad (12)$$

Here, the sign of f_d depends on the radial-motion direction, and a positive f_d is assumed for notational simplicity.

The communication data are embedded in the time variation of the tag reflection coefficient. For a modulation component at $f_{m,k}$, the reflection coefficient can be expressed as

$$\Gamma(t) = \Gamma_0 + \beta_k \cos(2\pi f_{m,k}t + \theta_k), \quad (13)$$

where Γ_0 is the static reflection component, β_k is the modulation amplitude, $f_{m,k}$ is the k -th tag modulation frequency, and θ_k is the corresponding modulation phase.

Substituting $\Gamma(t)$ into the dechirped signal gives

$$\begin{aligned} r_b(t) = & A\Gamma_0 \cos(2\pi(f_b + f_d)t + \phi_0) \\ & + \frac{A\beta_k}{2} \cos(2\pi(f_b + f_d + f_{m,k})t + \phi_0 + \theta_k) \\ & + \frac{A\beta_k}{2} \cos(2\pi(f_b + f_d - f_{m,k})t + \phi_0 - \theta_k). \end{aligned} \quad (14)$$

This result shows that the tag modulation produces sideband components around the range-Doppler beat frequency ($f_b + f_d$). Thus, the received frequency components are

$$f_{rx,k}^+ = f_b + f_d + f_{m,k}, \quad (15)$$

$$f_{rx,k}^- = f_b + f_d - f_{m,k}. \quad (16)$$

If the upper modulation sideband is used for communication, the received communication-bearing component can be simply represented as

$$f_{rx,k} = f_b + f_d + f_{m,k}. \quad (17)$$

Here, the tag modulation shifts the range-Doppler beat component by $f_{m,k}$, enabling communication-data recovery from the same dechirped baseband signal.

D. Key Insight of ISABC Systems

From the unified signal model, ISABC systems reuse the radar waveform for communication while simultaneously performing sensing. The tag does not generate RF signals but modulates the incident waveform, enabling ultra-low-power operation. Range information is obtained from the beat frequency, velocity information from the Doppler frequency, and communication data from modulation frequencies. As a result, sensing and communication share the same waveform, spectrum, and hardware infrastructure.

This unified signal model forms the theoretical foundation of ISABC systems and explains how passive tags can simultaneously support localization, sensing, and data communication.

III. SENSING BACKSCATTER AND COMMUNICATION BACKSCATTER SYSTEMS

Sensing backscatter systems and communication backscatter systems have historically been developed in parallel but largely independent research directions. Sensing backscatter systems primarily focus on localization, tracking, and environmental sensing using radar signals and passive tags, while communication backscatter systems focus on data transmission, modulation techniques, and network architectures for passive wireless communication. Despite these different research objectives, both systems rely on the same physical principle, namely modulation of reflected electromagnetic waves through impedance switching. This observation naturally leads to ISABC systems, which will be discussed in the next section. Before discussing integrated systems, this section reviews representative sensing backscatter systems and communication backscatter systems.

TABLE I
SENSING BACKSCATTER SYSTEMS.

Work	Frequency / Bandwidth	Range	Sensing / Localization Performance	Supported Function
SuperSight [4]	77 GHz FMCW radar, 4 GHz BW	8 m, measured in indoor NLOS scenarios with non-penetrable blockages and specular reflectors	Sub-cm 3D NLOS localization; position accuracy up to 5.7 mm, 5.5 mm, and 7.7 mm in x , y , and z at 8 m	NLOS localization
R-fiducial [6]	24–24.3 GHz FMCW radar	25 m, measured for traffic-infrastructure detection; 100% detection up to 25 m with 120° FoV	Median range error of 0.1 m and median angle error of 6°	Sensing / localization
Millimetro [7]	24 GHz FMCW radar, 250 MHz BW; conceptually extendable to 77 GHz	100 m, measured in LOS indoor/outdoor scenarios; >90% detection rate up to 100 m	Median localization accuracy of 15 cm over 100 m; median range error of 10–20 cm up to 100 m	Localization

A. Sensing Backscatter Systems

Sensing backscatter systems use radar signals and passive backscatter tags to perform localization, tracking, and environmental sensing. In these systems, the reader transmits radar signals such as FMCW chirps, and passive tags reflect and modulate the signal. The reader then processes the received signal using range fast Fourier transform (FFT), Doppler FFT, or angle estimation techniques to estimate the position and motion of the tags.

One of the representative sensing backscatter systems is Hawkeye, which demonstrates long-range localization using backscatter tags. Hawkeye achieves a median localization accuracy of approximately 7 mm at a range of 160 m by leveraging FMCW radar and backscatter tags that modulate the reflected signal with specific frequency signatures. The system distinguishes multiple tags by assigning different modulation frequencies to each tag, which appear as separate peaks in the frequency domain after FFT processing [5].

R-fiducial uses FMCW radar and backscatter tags to distinguish different tags using both range FFT and Doppler FFT. Each tag modulates the reflected signal with a unique modulation pattern, and the radar system separates the tags by analyzing the frequency components in both range and Doppler domains [6]. Similarly, Millimetro uses Doppler FFT to distinguish tags based on their modulation frequencies in the backscattered FMCW signal. The system demonstrates that multiple backscatter tags can be separated in the Doppler domain even when they are located at similar ranges [7]. Retroreflective radar readable road sign (RoS) introduces polarization-based backscatter tags that can be detected reliably in roadway environments. Instead of relying solely on frequency modulation, RoS uses the geometric layout of the tag and polarization rotation in the reflection pattern to improve tag detection and robustness in practical environments [8].

The sensing range in Table I should be interpreted together with the channel condition and the tag/radar architecture. SuperSight focuses on indoor NLOS localization, where the signal reaches the tag through specular reflections rather than a direct LOS path. Therefore, its reported 8 m range is not a long-range detection target, but a design tradeoff for achieving sub-cm 3D localization accuracy under NLOS multipath conditions. In contrast, R-fiducial reports a shorter measured range of 25 m because it targets low-latency and wide-FoV traffic-infrastructure detection rather than narrow-beam long-range localization; however, its range can be extended through chirp combining and higher effective radar link budget. Millimetro

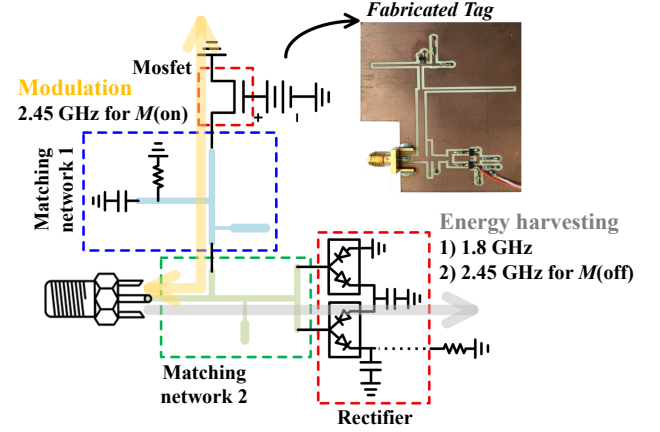


Fig. 5. Self-sustainable backscatter modulation scheme designing the combination of the dual-band rectifier and the conventional backscatter modulator. [36]

achieves long-range localization up to 100 m by assuming a mainly LOS radar-to-tag path and by using a Van-Atta-array-based retro-reflective tag, which concentrates the incident radar energy back toward the reader. Overall, the different ranges mainly reflect the tradeoff among EIRP, tag RCS/retro-reflective gain, antenna field of view, coherent integration time, and LOS/NLOS channel conditions.

B. Communication Backscatter Systems

In contrast to sensing backscatter systems, communication backscatter systems focus on wireless data transmission using passive backscatter tags. The main research objectives in communication backscatter include increasing data rate, improving communication range, designing efficient modulation schemes, and developing network architectures for passive communication systems.

One of the early communication backscatter systems is the backscatter wireless sensor network with wireless power transfer, which proposes a system that combines backscatter communication with wireless power transfer using two different frequencies. As shown in Fig. 5, one frequency is used for energy harvesting, while the other frequency is used for backscatter communication, enabling continuous operation of passive sensor nodes without batteries [36]–[39]. A key feature of this work is that the rectifier and backscatter paths are co-designed so that the node can continue harvesting energy even during communication, thereby avoiding the severe duty-cycle limitation of conventional passive tags. This architecture established an important foundation for self-sustained backscatter

TABLE II
COMMUNICATION BACKSCATTER SYSTEMS AND MODULATORS.

Work	Frequency / Bandwidth	Range / Test Scenario	Data Rate	Modulation / EIRP or Excitation Condition
mmTag [16]	24 GHz ISM band, 250 MHz available BW	4.6 m, 8 m, and 14 m measured communication range	1 Gbit/s at 4.6 m; 100 Mbit/s at 8 m; 10–15 Mbit/s at 14 m	ASK backscatter; AP transmit power of 12 dBm including antenna gain
Ultrafast backscatter modulator [20]	Backscatter: 2.45 GHz; WPT: 1.7 GHz	Modulator-level characterization; wireless link range N/R	960 Mbit/s with 8.37% EVM; 61.5 fJ/bit	16-QAM; EIRP N/R; WPT input power of 5 dBm at 1.7 GHz
Printed mmWave modulator [21]	24–28 GHz	0.5 m laboratory backscatter link	1 Gbit/s BPSK; 1 Gbit/s QPSK; 2 Gbit/s 16-QAM; 0.17 pJ/bit	BPSK, QPSK, and 16-QAM; EIRP N/R; 24.5-GHz CW illumination with high-gain horn antennas
BiCMOS mmWave backscatter modulator [26]	20–28 GHz	On-wafer / modulator characterization; wireless link range N/R	Static EVM characterization for 16/32/64-QAM; multigigabit-class link potential	16-QAM, 32-QAM, and 64-QAM; EIRP N/R; input power characterized at 0, –10, and –20 dBm
OFDM backscatter modulator [19]	2.45 GHz ISM band	1 m bistatic over-the-air validation	1.25 Mbit/s using five BPSK subcarriers; 195 kbit/s OTA validation	Five-subcarrier BPSK OFDM backscatter; EIRP N/R; 0-dBm CW source with RF amplification and 10-dBi antenna
IQ impedance LoRa backscatter [18]	2.45 GHz; LoRa-like CSS; 125 kHz available BW	10 m LoS, 7.5 m NLoS, and 10 m wall-obstructed indoor scenarios	Raw LoRa-like rate of approximately 6.8 kbit/s	LoRa-compatible CSS via IQ impedance modulation; EIRP N/R; required carrier input power of –40 dBm (LoS), –25 dBm (NLoS), and –26 dBm (wall) for error-free transmission
FM Rider [22]	Ambient FM band, 88–108 MHz	58 m for maximum bit-rate case; 107 m maximum tag-to-reader range	333 bit/s at 58 m with BER 2.85%; 200 bit/s at 107 m with BER 6%	In-band 2FSK ambient FM backscatter; EIRP N/R; ambient FM station 34.5 km away with measured received FM power around –40 dBm
Four-PAM ambient FM backscatter [23]	Ambient FM broadcasting band; BW N/R	1 m tag-to-reader indoor proof-of-concept; FM station 34.5 km away	345 bit/s; 78.2 nJ/bit at 345 bit/s	4-PAM ambient FM backscatter; EIRP N/R; ambient FM carrier used instead of dedicated RF source

N/R: not reported.

nodes in which sensing, powering, and uplink transmission can be supported simultaneously.

As shown in Fig. 6(a), mmTag demonstrated that millimeter-wave backscatter links can support high-throughput and low-power communication through passive beam alignment and polarization-based self-interference suppression, thereby providing an important hardware and propagation foundation for later mmWave backscatter communication systems [16]. In this system, the backscatter node modulates and reflects the received signal while also changing the polarization and reflecting the signal back toward the direction of arrival, improving link efficiency and communication range. More specifically, mmTag adopts a Van Atta-type retro-directive structure so that the reflected beam automatically returns toward the access point without requiring active phased arrays or beam training at the tag. In addition, polarization conversion allows the access point to separate the weak backscatter signal from the strong transmitted signal, which is essential for practical full-duplex-like operation at mmWave frequencies. The performance trend in Fig. 6(b) further illustrates how the achievable data rate scales with link distance and received power, highlighting the practical trade-off between throughput and range in mmWave backscatter systems.

Hardware-efficient orthogonal frequency division multiplexing (OFDM) backscatter communication systems have also been proposed, where all-digital architectures are used to implement OFDM backscatter modulators using digital complementary metal oxide semiconductor (CMOS) RF switches [19]. These systems demonstrate that high-order modulation schemes and multi-carrier communication can be implemented using backscatter modulation without requiring active RF transmitters. Their main contribution is to show that digitally controlled discrete impedances can approximate single-sideband OFDM modulation, enabling multipath-robust and spectrally flexible uplink with very low modulator power.

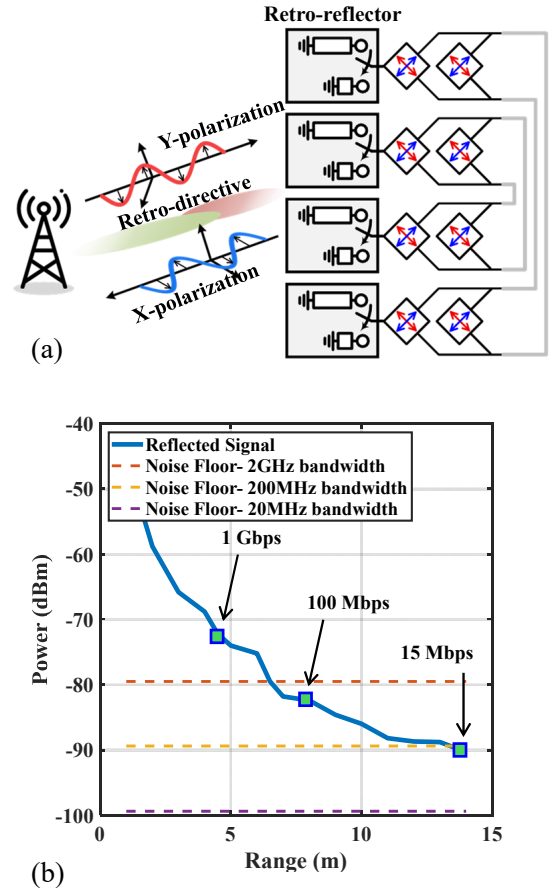


Fig. 6. A mmWave backscatter communication platform with passive beam alignment and polarization-based self-interference suppression: (a) system model and (b) measured communication performance under different ranges of the mmTag [16].

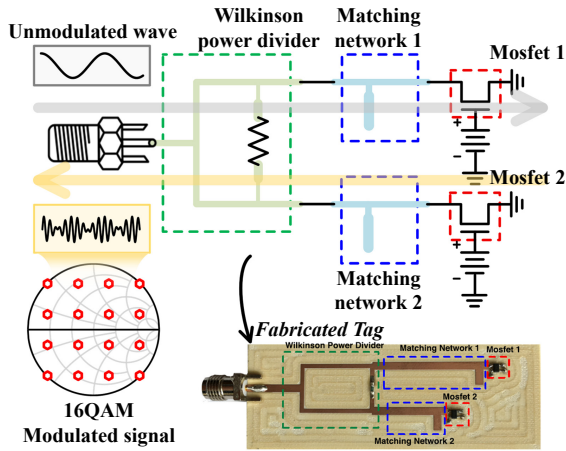


Fig. 7. 16-QAM backscatter modulation scheme using the Wilkinson power divider, which is effective to minimize the number of the transistors. [17]

This line of work is particularly important because it extends backscatter from simple single-carrier on-off keying (OOK)/frequency shift keying (FSK) links toward configurable multicarrier physical layers that are compatible with modern wireless networking concepts.

As shown in Fig. 7, other communication backscatter research focuses on higher-order modulation schemes such as quadrature amplitude modulation (QAM) backscatter modulation, multi-level impedance modulation, and frequency-shift backscatter modulation to improve spectral efficiency and data rate [17], [18], [23]. These systems treat the passive tag as a communication transmitter that encodes digital information onto the reflected signal. For example, QAM backscatter modulators engineer multiple reflection-coefficient states so that both amplitude and phase can be controlled, enabling more than one bit per symbol and therefore much higher spectral efficiency than binary switching. Similarly, in-phase and quadrature (IQ) impedance modulation and chirp- or frequency-based backscatter schemes show that passive nodes can synthesize more advanced waveforms such as LoRa-like chirps or multilevel ambient-FM symbols, broadening the design space of passive communication beyond conventional RFID-style modulation.

Table II summarizes the communication-oriented backscatter systems from the viewpoint of carrier frequency, range, data rate, and modulation method. The comparison shows that high-data-rate backscatter is mainly achieved either by using wide mmWave bandwidth or by employing high-order modulation in a controlled link budget. For example, mmTag and the printed mmWave modulator demonstrate Gbit/s-class communication at short-to-moderate ranges by using directional mmWave links and ASK/QAM-type modulation [16], [21]. In contrast, low-frequency or ambient-source systems, such as FM Rider and four-PAM ambient FM backscatter, sacrifice data rate to obtain longer range or infrastructure-free operation by reusing broadcast FM signals [22], [23]. Other works, including OFDM backscatter and LoRa-compatible IQ impedance modulation, focus on improving robustness

or spectral efficiency rather than only maximizing raw data rate [18], [19], [27], [28]. Overall, communication-oriented backscatter systems exhibit a clear tradeoff among data rate, operating range, modulation complexity, excitation power, and channel robustness.

C. Comparison Between Sensing Backscatter and Communication Backscatter

Although sensing backscatter and communication backscatter systems have been developed independently, they share the same underlying physical principle: both systems modulate the reflection coefficient of a passive tag to encode information onto the reflected electromagnetic wave. The main difference lies in how the reflected signal is processed at the receiver.

In sensing backscatter systems, the receiver processes the signal using radar signal processing techniques such as range FFT, Doppler FFT, and angle estimation to extract localization and motion information. In communication backscatter systems, the receiver demodulates the signal to extract digital communication data.

In summary, sensing backscatter and communication backscatter systems are not fundamentally distinct but rather represent different signal processing strategies applied to the same reflected waveform. This observation directly motivates ISABC, which unifies both functions within a single passive backscatter architecture. This relationship is summarized in Table III.

IV. INTEGRATED SENSING AND BACKSCATTER COMMUNICATION SYSTEMS

A. Concept of ISABC

ISABC systems aim to simultaneously perform localization, sensing, and communication using passive backscatter devices and radar waveforms. Unlike conventional backscatter communication systems that focus only on data transmission and sensing backscatter systems that focus only on localization or environmental sensing, ISABC systems combine both functionalities into a unified signal processing framework.

The key idea of ISABC is that the backscattered signal contains multiple frequency components corresponding to range, velocity, and communication data. When an FMCW radar waveform is transmitted, the reflected signal from a passive backscatter tag produces a beat frequency component that is related to the target range. If the tag modulates its reflection coefficient, additional frequency components appear around the beat frequency. These modulation sidebands can be used to encode communication data, while the beat frequency still provides range information.

Therefore, sensing and communication can be performed simultaneously by analyzing the frequency components of the received signal. Range information is obtained from the beat frequency, velocity information from the Doppler frequency, and communication data from modulation frequencies. This unified signal processing framework enables passive backscatter devices to support localization, sensing, and communication simultaneously without active RF transmitters.

TABLE III
CLASSIFICATION OF BACKSCATTER SYSTEMS ACCORDING TO FUNCTIONALITY AND SIGNAL PROCESSING.

System Type	Objective	Signal Processing
Sensing Backscatter	Localization, tracking, and environmental sensing	Range FFT, Doppler FFT, and angle estimation
Communication Backscatter	Data transmission between tag and reader/receiver	Demodulation and decoding for encoded FMCW signal from tag
Integrated Sensing and Backscatter Communication	Simultaneous localization, sensing, and communication	FFT-based sensing pipeline combined with demodulation

TABLE IV
REPRESENTATIVE INTEGRATED SENSING AND BACKSCATTER COMMUNICATION SYSTEMS.

Work	Frequency / Bandwidth	Range-Dependent Data Rate	Sensing / Localization Performance	Supported Function
Multichannel FMCW Backscatter [10]	2.45 GHz, 40 MHz bandwidth	10 kbit/s per sensor at 15 m and 33 m	Simultaneous ranging and telemetry uplink for two sensor nodes in a cluttered indoor environment	Sensing and uplink communication
UniScatter [11]	Wideband response from 24 to 77 GHz; demonstrated with 24, 60, and 77 GHz devices	20 bit/s tag modulation; around 30 m communication distance with SNR > 10 dB and raw BER < 5.7%	Wide angular coverage over 70° in 3D; compatible with heterogeneous radar and JCAS devices	Sensing and uplink communication
MilBack [15]	26.5–29.5 GHz, 3 GHz bandwidth	Downlink up to 36 Mbit/s; uplink 10 Mbit/s at 8 m and 40 Mbit/s at 6 m; maximum uplink rate of 160 Mbit/s	Distance error below 5 cm at 5 m and below 12 cm at 8 m; median/90th-percentile angle error of 1.1°/2.5°	Localization and two-way communication
BiScatter [9]	9 GHz with 1 GHz bandwidth; 24 GHz with 250 MHz bandwidth	Two-way communication with BER < 10 ⁻³ up to 7 m; current data rate of about 50–100 kbit/s	Centimeter-level tag localization accuracy up to 7 m using off-the-shelf FMCW radars	Sensing, localization, and two-way communication

Another important characteristic of ISABC systems is that the same hardware platform can be used for both sensing and communication. The radar transmitter, receiver, antennas, and signal processing blocks are shared between sensing and communication functions. This reduces system complexity, power consumption, and hardware cost, making ISABC systems suitable for large-scale IoT sensing networks and future 6G integrated sensing and communication systems.

B. Integrated Sensing and Backscatter Communication Systems Using FMCW Radar

Several representative studies have shown that an FMCW radar can be extended from a pure sensing platform into an ISABC system by allowing passive or semi-passive tags to modulate the reflected radar waveform. The key principle is that the FMCW signal already provides range information through the beat frequency after dechirping, while the tag embeds communication data by altering the spectral content of the reflected waveform through impedance switching, frequency shifting, or waveform-aware modulation. As a result, a single interrogating radar waveform can simultaneously reveal the tag position and recover its data stream.

One of the earliest representative examples is the multichannel FMCW backscatter system in [10]. In this work, the radar transmits standard FMCW chirps and each tag modulates the incident signal through a simple load-switching backscatter mechanism, as shown in Fig. 8(a). The sensing function is obtained from the conventional FMCW beat-frequency principle: after mixing the received echo with the transmitted chirp, the radar extracts a beat tone whose frequency is proportional to the round-trip delay, and hence to the tag range. Communication is superimposed on top of this process because the tag intentionally introduces modulation-induced spectral components on the reflected FMCW waveform. By

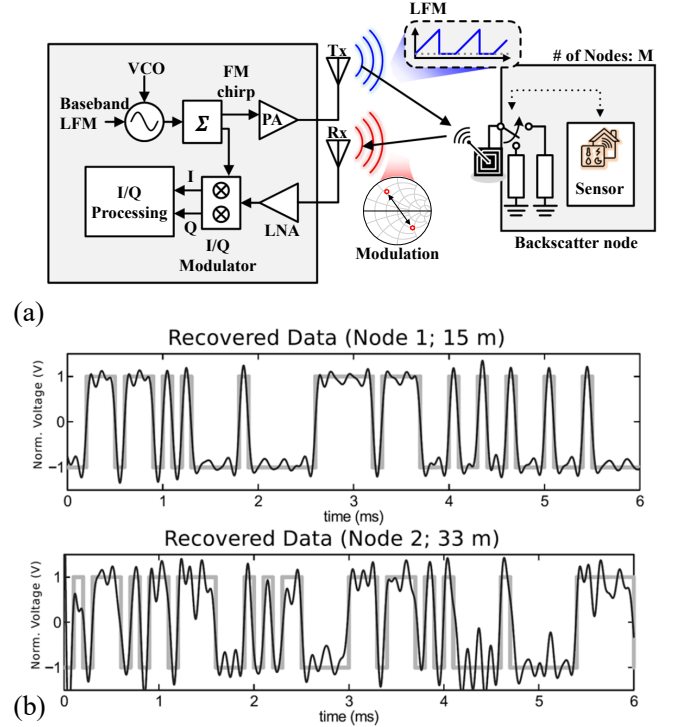


Fig. 8. Multichannel FMCW-based ISABC system: (a) system model and (b) comparison between the recovered and transmitted data waveforms for two nodes at different ranges [10].

separating these components in the frequency domain, the radar can decode the uplink data while preserving range estimation. The recovered waveforms in Fig. 8(b) further confirm that reliable data demodulation is maintained over different ranges, demonstrating the robustness of simultaneous

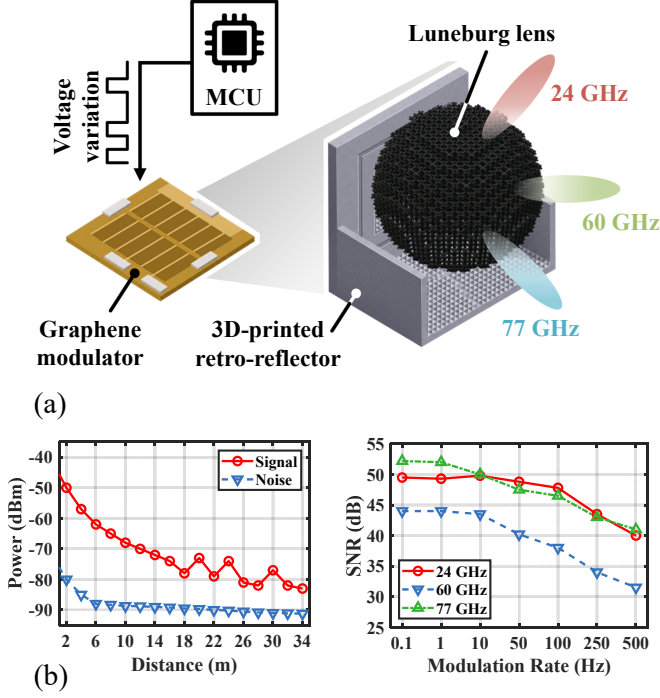


Fig. 9. An advanced backscatter tag based on the metamaterial lens and graphene modulator for mmWave ISABC system: (a) UniScatter structure backscattering FMCW signals within the multi-frequency and (b) localization and communication performance of UniScatter [11].

sensing and communication in practical scenarios. An important contribution of this work is that multiple tags can be separated by assigning different modulation frequencies, so the same FMCW radar can simultaneously perform ranging and multi-channel telemetry from distributed passive sensor nodes.

A more advanced millimeter-wave realization is UniScatter in [11], which addresses a different but equally important limitation: the hardware scalability of radar backscatter tags across wide bandwidth and wide angular coverage. UniScatter consists of two tightly coupled hardware components, namely a graphene-based modulator and a Luneburg-lens-based retroreflector, as shown in Fig. 9(a). The communication function is achieved by changing the reflection coefficient of the graphene metasurface, so that the reflected radar signal carries modulation sidebands or coded spectral signatures. The sensing function remains radar-centric: an FMCW radar estimates range from the beat-frequency spectrum while also identifying the tag through the imposed modulation pattern. The main innovation, however, lies in the tag architecture itself. Unlike conventional resonant antenna tags that work only over a narrow band and limited angular region, UniScatter maintains strong and consistent backscatter from 24 GHz to 77 GHz and over a wide 3D field of view. The Luneburg lens improves retroreflection and radar cross-section, while the graphene modulator preserves controllable reflection over a broad frequency range. The measurement results in Fig. 9(b) further validate that the tag maintains a clear signal-to-noise margin and stable performance across distance, frequency,

and modulation rate, confirming its broadband and angle-robust operation. Therefore, UniScatter shows that ISABC performance is not determined only by the waveform and digital signal processing (DSP), but also by how the tag hardware is engineered to support broadband, angle-robust joint sensing and communication.

Another important milestone is MilBack, which expands the scope of integrated backscatter systems from uplink-only telemetry to uplink, downlink, and localization in the same mmWave platform [15]. As shown in Fig. 10(a), MilBack introduces a passive node architecture that can operate in both absorptive and reflective modes. In the reflective mode, the node backscatters the incident mmWave signal toward the access point, enabling uplink communication. In the absorptive mode, the node receives energy from the interrogating signal and extracts downlink information. Localization is enabled by exploiting the mmWave signal structure and the directional behavior of the tag, so that the access point can estimate position and orientation from the reflected spectral response. For communication, MilBack proposes a low-complexity modulation strategy that avoids conventional IQ demodulation overhead and instead uses frequency-based symbol formation that is more suitable for passive mmWave nodes, as shown in Fig. 10(a). The corresponding performance evaluation in Fig. 10(b) further demonstrates the achievable communication reliability and localization accuracy as a function of distance, highlighting the practical feasibility of the proposed integrated architecture. Its significance lies in showing that the backscatter tag can be more than a simple reflective endpoint: by co-designing node structure, beam behavior, and modulation method, the same hardware can support localization together with two-way communication.

The most complete realization of this idea is BiScatter in [9], which explicitly targets simultaneous uplink communication, downlink communication, radar sensing, and localization using off-the-shelf FMCW radars. As shown in Fig. 11, the key downlink innovation is chirp-slope-shift keying (CSSK), where information is embedded into the radar waveform itself by varying the slope of the FMCW chirp. A passive tag equipped with low-power differential circuitry and delay lines observes the resulting beat-frequency change and decodes the downlink message without requiring an active RF receiver, as shown in Fig. 11. At the same time, the tag performs uplink communication by modulating its reflected signal, so the radar can recover the tag's data from the backscatter spectrum while still performing conventional FMCW range estimation. In this sense, sensing and communication are not merely overlaid, but are jointly designed through the waveform, tag circuit, and radar receiver processing chain. BiScatter is particularly important because it demonstrates that a commercial FMCW radar can be transformed into a full ISABC node without abandoning radar compatibility, while achieving centimeter-level localization and low bit error rate (BER) two-way communication.

Communication can also be integrated into radar operation by modifying the radar waveform parameters themselves rather than only changing the reflection coefficient of the tag. This idea is exemplified by the BiScatter framework, where

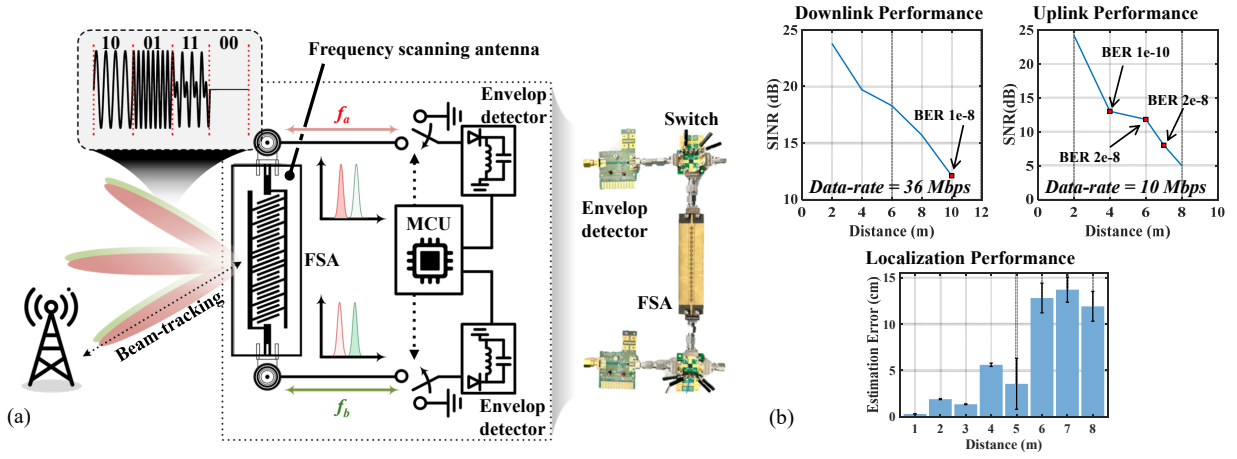


Fig. 10. A mmWave ISABC system for two-way communication and localization: (a) system model with an inset showing the node implementation and (b) localization and communication performance of MilBack [15].

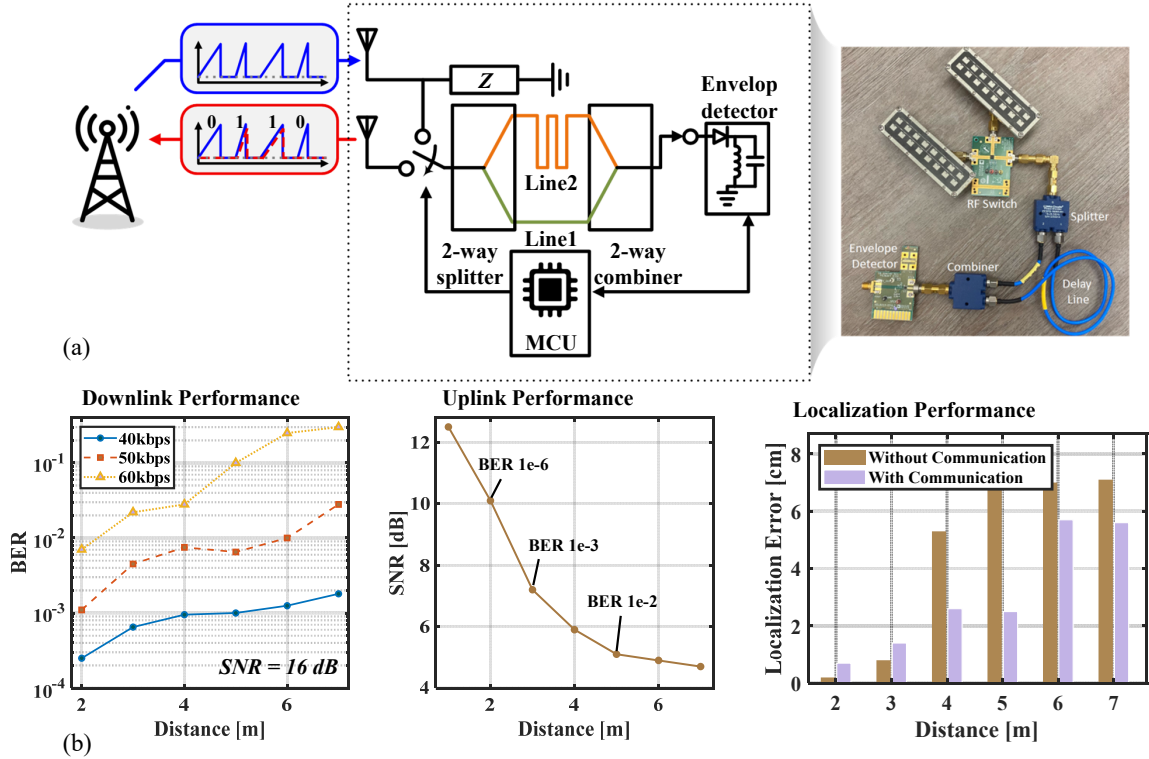


Fig. 11. A mmWave ISABC system based on CSSK and low-power downlink decoding concept: (a) system model with an inset showing the node implementation and (b) localization and communication performance of BiScatter [9].

the downlink is carried by chirp-slope variation and the tag decodes it through passive delay-based frequency interpretation. This is an important conceptual step because it expands the ISABC design space beyond conventional impedance-switching backscatter. In earlier systems, the tag primarily acts as a modulated reflector, whereas in this type of architecture the radar waveform itself becomes a communication-bearing signal for the tag. As a result, the overall system supports a richer interaction model in which the radar not only senses the tag and receives its uplink response, but also configures or controls the tag through embedded downlink signaling.

Taken together, these studies reveal a clear technical progression. The early FMCW backscatter work in [10] established the basic principle that beat-frequency processing can support simultaneous ranging and uplink telemetry. UniScatter then showed that practical mmWave ISABC requires a new tag hardware paradigm with wideband modulation and robust retroreflection [11]. MilBack extended the architecture toward localization-aware and bidirectional backscatter networking [15]. Finally, BiScatter unified radar sensing, localization, uplink, and downlink within a single FMCW-compatible framework [9]. Thus, the evolution of ISABC can be interpreted as a

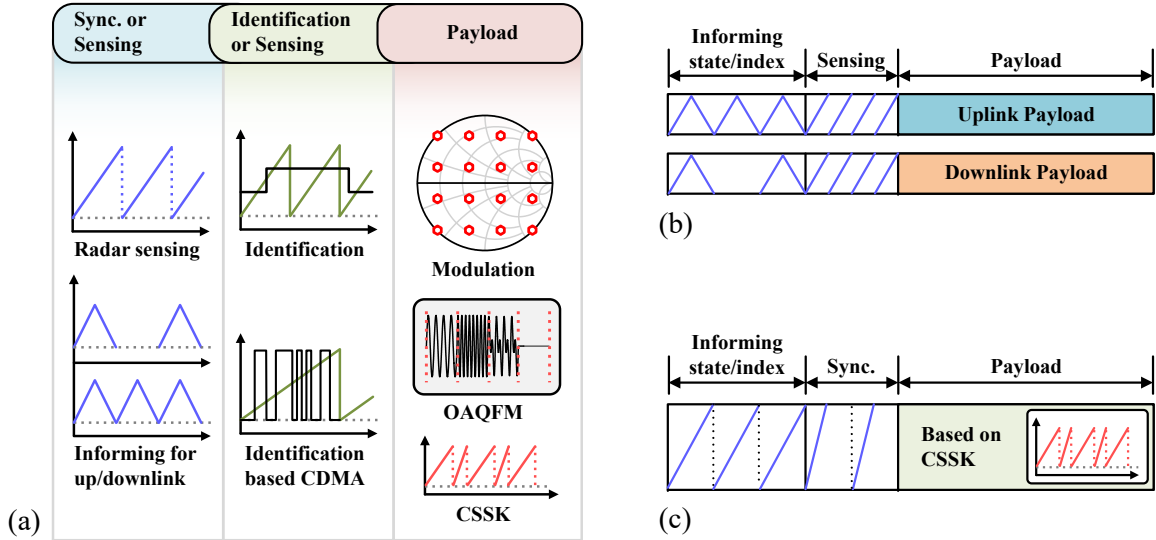


Fig. 12. Unified functional packet structure for complex ISABC situations of two-way communication and existence of multi-tag: (a) description of the general packet structure, (b) timing-aware packet structure [9], and (c) link-state informing packet structure [15].

progression from range-plus-uplink sensing tags to broadband mmWave reflective platforms, and finally to fully interactive radar-backscatter nodes that are jointly designed across waveform, hardware, and signal processing layers.

Table IV further summarizes this progression from a system-level performance perspective. The multichannel FMCW backscatter system demonstrates the baseline feasibility of simultaneous ranging and uplink communication, but its data rate remains limited to the kbit/s level because the design prioritizes compatibility with conventional FMCW radar processing [10]. UniScatter extends this concept to a much wider mmWave frequency range from 24 to 77 GHz, showing that broadband and angle-robust tag hardware is essential for practical radar-backscatter deployment, although its communication rate is still low because the emphasis is on robust tag design based on the meta-lens and the low-powered modulation across ultrawide bandwidth [11]. MilBack shifts the design target toward a networked mmWave backscatter node by supporting uplink, downlink, and localization with Mbit/s-class communication at several-meter ranges [15]. BiScatter, in contrast, sacrifices data rate for stronger FMCW-radar compatibility and low-power passive downlink decoding, while still achieving centimeter-level localization and reliable two-way communication up to 7 m [9]. Therefore, the comparison in Table IV shows that ISABC systems are not optimized along a single axis; instead, their performance is determined by the joint tradeoff among radar compatibility, communication directionality, data rate, localization accuracy, operating range, and tag hardware complexity.

C. Design Insights for ISABC Systems

The surveyed literature suggests that the evolution from sensing backscatter to ISABC is not merely a progression in modulation complexity, but a transition in how the reflected radar waveform is functionally organized. In early sensing

backscatter systems, the passive tag primarily acted as a radar-visible reflector with an embedded signature for identification. In more advanced ISABC systems, the same reflected waveform is further exploited to carry telemetry, uplink data, and in some cases even downlink interaction. From this perspective, as shown in Fig. 12(a), the most important design insight is that modern ISABC can be interpreted as a structured coexistence of three functions within a single radar-backscatter interaction: detection/synchronization, tag identification, and payload communication.

This interpretation becomes clear when comparing representative sensing backscatter systems with ISABC-oriented systems. In sensing-oriented designs, the tag modulation is rarely used for high-rate communication; instead, it is mainly introduced to make the tag distinguishable from clutter and from other tags. For example, the multichannel FMCW backscatter system in [10] uses a low-rate preamble before the sensor data to improve range estimation and distinguish the node response from stationary clutter. It also employs range-dependent subcarriers for data recovery. Therefore, although the system is not described using the terminology of a structured radar-backscatter frame, it already includes its essential elements. Likewise, R-fiducial assigns each tag a unique spreading code and jointly estimates identity and distance through correlation over range bins, effectively treating the reflected waveform as an identification-bearing radar response rather than a pure communication signal [6]. Millimetro similarly separates multiple tags by assigning different modulation frequencies, so that the switching signatures appear as distinguishable components in the range-Doppler profile [7]. In all of these systems, the modulation is not only a sensing aid but also an implicit label attached to the radar target.

In contrast, ISABC systems extend this role of modulation from identification to information transfer. The basic FMCW backscatter architecture in [10] already demonstrated that the dechirped signal can simultaneously preserve range informa-

tion and recover telemetry from the backscatter spectrum. Later systems strengthened this principle in different directions. UniScatter showed that robust ISABC operation depends not only on waveform processing but also on tag hardware that sustains controllable reflection over a broad frequency range and wide angular coverage, using a graphene-based modulator and a Luneburg-lens retroreflector for wideband radar-backscatter interaction [11]. More broadly, the ISABC concept formalized in [12] clarified that the passive tag is no longer treated as a mere radar target, but as a sensing target and information source simultaneously. This changes the role of the reflected waveform: it must support radar observables such as range, velocity, or angle, while also carrying information intended for a user or infrastructure node [12].

Taken together, these studies suggest that an effective IS-ABC architecture should not be understood as simple “data modulation on top of FMCW,” but rather as a layered waveform interaction in which different portions or features of the reflected signal serve different system functions. A useful abstraction is to interpret the backscatter response in a frame-oriented manner: a known or structured component supports detection and synchronization; a dedicated signature-bearing component supports tag identification and multi-tag separation; and a payload-bearing component conveys sensor data or communication symbols. Importantly, this does not necessarily require explicit packetization in the conventional digital-communications sense. Instead, it means that the radar-backscatter system should be designed so that these three functions are separable in processing, whether through time segmentation, modulation-frequency allocation, code-domain separation, chirp-to-chirp patterning, or waveform-parameter embedding.

This viewpoint leads to several concrete design implications. First, identification should be treated as a first-class design objective rather than a byproduct of communication. The sensing backscatter literature shows that reliable tag identification is fundamental for clutter suppression, multi-tag operation, and low-latency detection, especially in radar-centric environments where many passive reflectors coexist [6], [7]. Second, as shown in Fig. 12(b), communication modulation must be designed with awareness of radar observables, such as link states (uplink/downlink) and synchronization [9], [15]. Since the dechirped FMCW signal already maps propagation delay into beat frequency, additional modulation should be placed such that it remains distinguishable from range- and Doppler-related components rather than corrupting them [10]. Third, the hardware of the tag must be co-designed with this signal structure, as shown in Fig. 12(c). As UniScatter illustrates, if the reflection coefficient, radar cross-section, or angular response is unstable across frequency and direction, then neither identification nor data recovery can remain robust in practical deployments [11]. Finally, the most scalable future ISABC systems will likely be those that explicitly organize sensing, identification, and communication as separate but coordinated layers of the reflected waveform, instead of assuming that a single undifferentiated modulation mechanism can serve all functions equally well.

Therefore, one of the key lessons from the surveyed works

is that the path from backscatter sensing to full ISABC is not simply a shift from localization to communication, but a progression toward functionally structured radar-backscatter interactions. Early systems established how tags can be made visible and distinguishable to radar. ISABC systems then showed how those distinguishable reflections can also transport data. The next logical step is to systematically design radar-backscatter waveforms and tag responses so that synchronization, identification, and communication are jointly supported within a unified framework. This perspective provides a clearer architectural basis for future ISABC systems intended for smart infrastructure, industrial sensing, autonomous mobility, and 6G sensing-aware IoT networks.

V. TOWARD ROBUST HIGH-RATE ISABC: MODULATION, SPATIAL EXPANSION, AND REFLECTIVE HARDWARE

The representative ISABC systems discussed so far establish the basic feasibility of using a single interrogating waveform for both radar sensing and backscatter communication. However, most existing demonstrations still remain constrained by limited payload rate, weak backscatter link budget, narrow angular coverage, or poor multi-tag scalability. From this perspective, the next stage of ISABC development is not merely to combine sensing and communication in the same platform, but to improve how efficiently information is embedded into the reflected waveform, how robustly it survives propagation, and how flexibly the tag hardware can shape the reflection itself [10]–[12].

A useful interpretation of the recent literature is that future ISABC performance will be determined by three tightly coupled design axes: waveform-efficient modulation, spatial expansion through MIMO and diversity, and reflection engineering through advanced modulator and metasurface hardware. These are not isolated research directions. Rather, they jointly determine whether ISABC can evolve from proof-of-concept radar tags into scalable platforms for dense sensing networks, smart infrastructure, and 6G sensing-aware IoT [2], [11], [12].

A. From Simple Impedance Switching to Waveform-Efficient Modulation

A primary bottleneck of early backscatter communication is that the tag modulation mechanism is intentionally simple. Binary load switching based on OOK, amplitude shift keying (ASK), binary phase shift keying (BPSK), or low-order FSK minimizes circuit complexity and power consumption, but it also restricts spectral efficiency and limits the amount of information that can be carried by the reflected waveform. This limitation becomes more critical in ISABC, where the same interrogation waveform must preserve radar observables such as range and Doppler while also carrying communication symbols [1], [2], [17]. As a result, improving the communication throughput of ISABC requires not only higher-order modulation at the tag, but also a receiver and frame structure that remain compatible with radar processing.

This is why the backscatter literature gradually moved toward higher-order modulation. One representative example is the quadrature-amplitude backscatter modulator of [17],

which uses a Wilkinson power divider, a phase-shifted branch, and transistor-based impedance control to realize M-QAM signaling with passive backscatter hardware. In parallel, multi-level amplitude modulation schemes such as 4-PAM were introduced to improve spectral efficiency in ambient backscatter by creating multiple amplitude states in the reflected signal [23]. Although these studies were not originally developed for radar-integrated systems, they are directly relevant to ISABC because they demonstrate that the reflected waveform need not be limited to binary tag signatures. Instead, the backscatter response can be engineered as a communication-bearing signal with controllable amplitude and phase states [17], [23], [25]. In this sense, quadrature phase shift keying (QPSK)- or QAM-based ISABC can be viewed as a natural extension of recent progress in waveform-efficient backscatter design.

At the same time, integrating QPSK or 16QAM into an FMCW-based ISABC platform is not a straightforward extension of conventional backscatter modulation. After de-chirp processing, the communication symbols appear on top of range-dependent beat components rather than as an isolated baseband waveform. Therefore, a practical receiver must jointly perform range estimation, beat-tone or subcarrier isolation, range-dependent subcarrier wipe-off, and carrier/phase recovery before reliable QPSK or QAM demodulation becomes possible. This is fundamentally more demanding than binary detection because residual ranging error, clutter leakage, and phase distortion directly degrade the recovered constellation.

For this reason, waveform-efficient ISABC also requires a more structured frame design. A practical packet should include a radar-friendly preamble for detection and coarse synchronization, a tag-ID or header field for node association, a pilot block for post-de-chirp phase and amplitude calibration, and a payload block carrying the higher-order symbols. Such a framing strategy is particularly important in distributed sensing scenarios, where multiple tags may occupy different ranges or partially overlapping returns. The preamble supports robust detection in the radar domain, while the header and pilot enable coherent symbol recovery in the communication domain.

Another important transition is the move from carrier-oriented modulation to waveform-aware modulation. Since many ISABC platforms rely on FMCW radar, modulation schemes that are compatible with chirp-like incident signals are more attractive than amplitude-only methods designed for simple CW excitation. In this regard, chirp spread spectrum backscatter and IQ impedance modulation are especially important. Belo *et al.* showed that an IQ impedance modulator can generate LoRa-like LFM chirps by properly varying the phase of the reflected wave rather than directly synthesizing frequency at the tag [18]. Their implementation uses two RF transistors and quadrature combining so that the reflection phase spans a full 360° on an approximately constant-VSWR circle, making chirp-compatible backscatter modulation feasible with low-power hardware [18]. Similarly, OOK chirp spread spectrum backscatter demonstrates that chirp-compatible backscatter can also be realized in wireless power transfer (WPT)-assisted passive platforms without relying on

a conventional active transmitter chain [40]. From an ISABC perspective, the importance of such approaches is not only that they can improve data rate, but also that they provide a natural framework for embedding communication symbols into radar waveforms that are already frequency-structured [18], [40].

Taken together, these developments suggest that the next step for ISABC is not simply to replace BPSK with QPSK or 16QAM, but to co-design the tag modulator, radar waveform, packet structure, and receiver processing as a unified system. Only through such joint design can higher-order backscatter modulation increase the data rate without undermining the ranging and sensing functions that distinguish ISABC from conventional backscatter communication.

B. Full-Duplex Modulator Hardware as the Enabler of Radar-Compatible Communication

A radar-compatible ISABC link differs fundamentally from a conventional one-way backscatter uplink. The access node must continuously emit a sensing waveform, receive the weak tag-modulated echo, and preserve radar observables such as range, Doppler, and angle. Therefore, the ISABC link should not be treated merely as a tag-to-reader telemetry channel. Instead, it requires hardware architecture that can support simultaneous illumination, reception, and modulation within the same time-frequency resource. In this sense, full-duplex operation becomes an enabling condition for ISABC, rather than merely a protocol-level option achieved by adding handshaking procedures or separating uplink and downlink through time-division duplexing [9], [12], [15].

The need for full-duplex capability originates from the intrinsic carrier-interference problem of in-band backscatter [41]. Since the desired backscatter response is a modulated version of the incident carrier, the receiver must detect an extremely weak reflected signal in the presence of a much stronger leakage or carrier component. This problem becomes more severe in mmWave ISABC, where directional beams are used to compensate for path loss, but transmitter-to-receiver leakage can still limit receiver gain and saturate the front-end. Consequently, a radar or access point interacting with a backscatter tag must either employ full-duplex cancellation hardware or introduce an equivalent mechanism that separates the desired tag response from the strong self-interference [16].

Early full-duplex backscatter work showed that instantaneous feedback can be realized with simple passive analog components, enabling collision termination, rate adaptation, and bit-level retransmission without using power-hungry cancellation circuits [42]. Network-level extensions such as time-hopping spread spectrum further demonstrated that full-duplex information transfer and wireless energy transfer can coexist in dense BackCom networks [43]. However, these approaches mainly address link-layer feedback or multiple-access interference. The technique cannot fully remove the physical-layer carrier-cancellation bottleneck required by radar-compatible backscatter, especially when the modulated waveform must remain available for sensing and the response of the backscatter tag must be recovered without sacrificing radar continuity.

Recent hardware-oriented studies therefore provide a more relevant direction for ISABC. Lotus revisits polarization mis-

match and demonstrates that carrier interference can be suppressed in the analog domain through a backscatter-specific antenna and tag design, avoiding the spectrum overhead of frequency shifting and the power/protocol overhead of conventional cancellation approaches [44]. Although Lotus is not a complete full-duplex transceiver in the conventional active-radio sense, it addresses the key physical bottleneck of in-band backscatter reception by exploiting polarization-domain separation between the direct carrier leakage and the desired tag-reflected echo. This enables carrier suppression without frequency shifting, thereby preserving spectrum efficiency while retaining the weak backscatter response.

Armstrong reported in [45] takes the next step toward a truly ISAC-ready full-duplex backscatter platform at mmWave frequencies. It introduces a full-duplex mmWave backscatter tag architecture based on regenerative uplink amplification and regenerative downlink reception, enabling simultaneous bidirectional communication at extended ranges with low-cost hardware [45]. This approach can particularly suggest a potential of full-duplex ISABC, where future tags should not only reflect a radar waveform for sensing, but also receive a modulated data while concurrently embedding backscattered signals into the reflected signal.

Taken together, these works suggest that the next stage of ISABC should move beyond half-duplex tag based on scheduling or purely protocol-level separation. Radar-compatible backscatter requires a full-duplex-capable hardware layer that can suppress carrier/self-interference, maintain continuous sensing illumination, and support bidirectional tag interaction. From this perspective, Lotus and Armstrong represent two complementary hardware paths: Lotus provides a low-overhead carrier-cancellation principle for backscatter reception, while Armstrong demonstrates a practical mmWave full-duplex tag design [44], [45]. Such full-duplex modulator hardware is therefore a key enabler for robust, high-rate, and sensing-compatible ISABC systems.

C. Spatial Expansion in ISABC: MIMO, Diversity, and Angular Multiplexing

If modulation determines how efficiently information is embedded into the reflected waveform, spatial expansion determines how reliably that information survives propagation and how cleanly it can be separated from other radar returns. As illustrated in the inset at the bottom of Fig. 13, spatial degrees of freedom are essential not only for improving communication robustness and diversity through multi-antenna backscatter techniques [27], but also for enabling polarization-domain multiplexing and dual-channel operation [28], as well as supporting angular selectivity and beam alignment for multi-tag separability [46]. From this perspective, MIMO is only one part of a broader design space that also includes polarization diversity, multibeam coverage, and angularly selective backscatter links.

On the communication side, backscatter systems have already shown that spatial processing can be exploited in multiple forms. MIMO backscatter architectures demonstrate that multiple reflecting elements and tag-side space-time coding can improve throughput and reduce error probability by

creating diversity-enhanced reflection paths. Versatile MIMO backscatter (VMscatter) is particularly relevant because it shows that a backscatter system can approach the diversity gain of conventional MIMO even though the reflected symbols are not orthogonal in the usual active-radio sense [27]. Likewise, the dual-channel polarization-diversity backscatter platform in [28] shows that polarization can serve as an additional spatial degree of freedom for high-speed passive links. Together, these results suggest that future ISABC systems can use spatial diversity not only to increase data rate, but also to improve the robustness of cascaded passive links that are inherently vulnerable to fading and weak backscatter returns [27], [28].

Recent lens-assisted millimeter-wave backscatter platforms further suggest that spatial expansion can contribute to higher throughput by enabling wide-angle multibeam operation in addition to conventional diversity gain. For example, lens-integrated mmID systems have demonstrated that broad angular coverage can be maintained together with high-order modulation and multi-gigabit backscatter transmission, while related fully passive and harvesting-oriented platforms indicate that such spatially extended operation can remain feasible even under passive or wireless-powered constraints [46], [47]. In parallel, sector-based beam-alignment backscatter has shown that spatially partitioned coverage can also be exploited for adaptive directional links, where different sectors are selectively aligned for passive communication while sustaining 16-ary modulation and up to 387 kb/s transmission over 20 m [29]. Although these works are not yet formulated as FMCW-based ISABC systems, they imply that future ISABC may increase data rate not only through higher-order modulation, but also through the use of multiple beams, wider angular sectors, and richer spatial channels.

At the same time, bringing spatial expansion into FMCW-based ISABC is not a direct extension of conventional MIMO or beam-steered communication systems. In radar-integrated backscatter, each reflected component must be resolved jointly in range, Doppler, and angle after de-chirp processing, rather than being treated as an isolated baseband spatial channel. This difficulty becomes even greater when multibeam or angularly multiplexed operation is introduced, because multiple reflected paths may coexist within the same observation interval and can overlap in the radar cube if their delay, Doppler, or angular signatures are not sufficiently separable. As a result, spatially expanded ISABC requires the joint design of array processing, tag modulation orthogonality, chirp-to-chirp coherence, beam-domain separability, and radar-cube association. Put differently, the benefit of multiple antennas, polarizations, channels, or beams is realized only when the reflected signals remain distinguishable in both the sensing domain and the communication domain. For this reason, the most promising role of spatial expansion in ISABC may be broader than spatial multiplexing alone: it can provide diversity gain for fragile passive links, angular resolution for localization, beam-level coverage extension, and multi-tag separability for distributed sensing, all within a common FMCW processing framework [27]–[29], [46], [47].

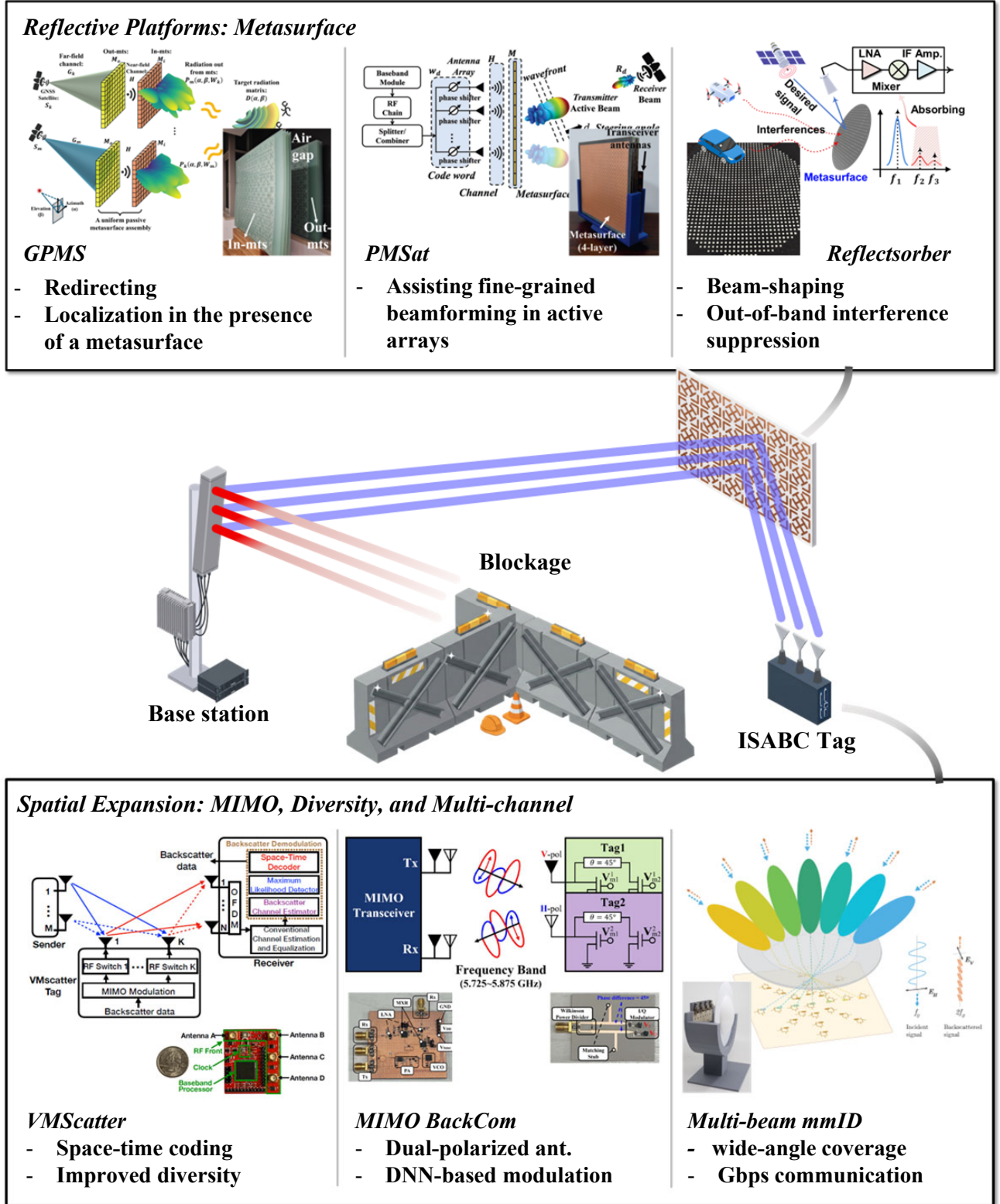


Fig. 13. Research directions for improving ISABC based on prior backscatter and wireless communication studies. Top: reflective metasurface platforms for mitigating blockage and path loss [30]–[32]. Bottom: spatial expansion through MIMO, diversity, and multi-beam techniques for improving data rate [27], [28], [46]. Representative prior works are addressed in each panel.

D. From Passive Tags to Reflective Platforms: Metasurfaces and Reflection Engineering

Even with higher-order modulation and spatial diversity, IS-ABC systems still face a fundamental bottleneck: the reflected

signal is often too weak, too angle-sensitive, or too unstable across frequency to sustain robust sensing and communication in realistic environments. This shifts attention from signal

processing alone to reflection engineering itself [3]. As illustrated in the top inset of Fig. 13, metasurface-based reflective platforms provide a concrete realization of this concept by enabling programmable control over wavefront redirection, beam shaping, and frequency-selective response.

More recent metasurface studies suggest that the role of reflection engineering is not limited to strengthening the tag response, but can be extended to creating or recovering useful propagation paths when the direct path is blocked, misaligned, or severely attenuated. At a broader system level, passive metasurface platforms further indicate that engineered reflective structures can actively create usable propagation paths rather than merely enhance local reradiation. PMSat shows that a passive metasurface, when jointly optimized with a small phased array, can realize fine-grained beam steering in the uplink and wide-angle focusing in the downlink, improving the signal-to-noise ratio (SNR) by 8.32 dB and 16.57 dB, respectively [30]. Likewise, GPMS demonstrates that passive metasurfaces can redirect weak GNSS signals from outdoors into indoor spaces and scatter them to enlarge coverage, thereby increasing the number of visible satellites and significantly improving indoor localization accuracy [31].

From an ISABC viewpoint, these results are important because they suggest that metasurfaces can be used to secure reflective paths in blockage-limited or shadowed environments, effectively extending the observable region of the radar-backscatter link rather than simply strengthening an already available line-of-sight reflection. This reflective freedom, however, comes with a new challenge for localization. In FMCW-based ISABC, an intentionally redirected or multi-hop reflection may appear as an additional propagation path with a different delay and angle, which can distort range-angle association if it is interpreted as a direct target return. In other words, the same metasurface that improves communication coverage may also create geometric ambiguity in sensing. Related mmWave backscatter localization studies already suggest both sides of this tradeoff: retroreflective structures can substantially improve detectability and range performance, but specular or redirected paths must be explicitly modeled rather than treated as nuisance-free echoes [4], [7], [11].

This perspective is also consistent with recent integrated metasurface concepts for wireless systems, where reflective surfaces are expected not only to steer useful signals but also to suppress or reshape undesired electromagnetic components through frequency-selective reflection and absorption [32]. For this reason, metasurface-assisted ISABC should be developed together with reflection-aware sensing algorithms. Future systems will likely need path labeling or environment-assisted inference to distinguish direct returns from engineered reflections, and may have to estimate not only the tag state but also the reflection state of the propagation channel. One possible direction is to treat the reflective surface as a known controllable node in the sensing loop, so that the radar does not interpret redirected energy as an unknown target but as a structured auxiliary path. Under this view, metasurfaces are not merely hardware extensions for stronger backscatter. They become channel-shaping components whose benefits for blockage mitigation, coverage extension, and signal en-

hancement must be balanced against the need for reliable localization and interpretable radar observables [30]–[32].

E. A Unified Perspective for Future ISABC Systems

Taken together, these developments point to a broader conclusion: future ISABC systems will likely advance not through a single dominant innovation, but through the joint evolution of modulation, spatial processing, and reflective hardware. Higher-order and waveform-aware modulation increases how much information can be embedded into a radar-compatible reflection [17], [18], [23], [40]. Advanced modulator circuits determine whether such modulation is physically realizable under passive or low-power constraints [21], [26], [36]. MIMO and polarization-diversity techniques enlarge the spatial degrees of freedom available for both sensing and communication [27], [28]. Metasurfaces and intelligent reflective tags further improve the link budget and reduce fading sensitivity by engineering the reflection process itself [11], [33].

Accordingly, the future of ISABC can be interpreted as a progression from simple modulated radar reflectors toward high-capacity reflective platforms. In early systems, the tag primarily served as a distinguishable radar target. In more advanced systems, it became a telemetry source. In the next generation, it is likely to become a programmable sensing-and-communication surface whose waveform response, spatial behavior, and reflection characteristics are jointly optimized. Such a progression is essential if ISABC is to support dense multi-tag sensing networks, robust operation in blocked or fading-prone environments, and higher-rate interaction with passive devices in future 6G and smart infrastructure scenarios [9], [11], [12], [27].

VI. CONCLUSION

Integrated sensing and backscatter communication is emerging as a compelling extension of backscatter technology from simple identification links to multifunctional passive wireless platforms. By allowing the same reflected waveform to carry both sensing observables and communication data, ISABC offers a unique path toward battery-free and low-complexity wireless nodes that can support localization, environmental awareness, and information transfer within a common hardware framework. Recent demonstrations based on FMCW radar, multichannel ranging, millimeter-wave retroreflective tags, and wideband reflective structures show that this concept is no longer only theoretical, but is steadily moving toward practical system realization.

At the same time, the field is still in an early stage. Fundamental challenges remain in overcoming double path loss, maintaining reliable communication without corrupting radar observables, separating multiple tags in shared spatial and spectral domains, and preserving performance under blockage, angular misalignment, and hardware constraints. These limitations suggest that future progress will depend not only on improved modulation or signal processing, but also on deeper co-design across waveform structure, synchronization, spatial processing, and electromagnetic hardware.

In that sense, the next generation of ISABC systems will likely move beyond conventional passive tags toward more intelligent reflective platforms. MIMO backscatter, polarization diversity, beam-aware architectures, and metasurface- or lens-assisted reflectors can provide new degrees of freedom to improve robustness, coverage, and spectral efficiency. As 6G research continues to push integrated sensing and communication into mainstream wireless system design, ISABC stands out as a particularly attractive solution for low-power IoT and smart infrastructure scenarios where battery replacement, hardware complexity, and deployment cost must be minimized. Its long-term significance lies in showing that sensing and communication need not be added as separate functions, but can instead be embedded together into the physics of reflection itself.

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